

PRATHEBA E

Data Science & Machine Learning Engineer

M.Tech Integrated · VIT Vellore · CGPA 8.5/10 · Research Co-Author · AWS Certified

CONTACT

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EDUCATION

M.Tech Integrated

Data Science (CS)
VIT Vellore
2023 – 2028 · CGPA 8.5

12th — CBSE

DAV Adambakkam, 2023

10th — CBSE

Asan Memorial, 2021

TECHNICAL SKILLS

Languages

Python · SQL · R · Java · C/C++

ML / DL

Scikit-learn · TensorFlow · Keras ·
Pandas · NumPy

NLP & LLMs

NLTK · SpaCy · CoT Prompting ·
LLM APIs

Databases

MySQL · Git · Linux · Jupyter ·
Colab

Concepts

ML · Image Processing · NLP ·
Time-Series · Data Viz

CERTIFICATIONS

AWS VPC Project Coursera · May 2026
Intro to Machine Learning Kaggle ·
May 2026
Pandas & Python Kaggle · May 2026
ICAT — CIT-P-3326004 iCAT · May
2026
Cyber Job Simulation Deloitte / Forage
· Aug 2025
Google Gemini Quiz Google GSA · Oct
2025
AI & DS Workshop NoviTech R&D ·
Sep 2024

PROFILE

Third-year M.Tech Integrated Data Science student at VIT Vellore (CGPA 8.5/10) with end-to-end ML project experience across NLP, image processing, and time-series domains. Co-authoring research on LLM-based drug interpretation using Chain-of-Thought prompting. Two-time certificate holder from ISOI VIT Chapter. Seeking a Data Science / ML internship to apply data-driven solutions to real-world problems.

PROJECTS

Drug Analysis Using MySQL · Python · MySQL · Image Processing *Team Project*

- Designed relational DB schema to store and query 500+ pharmaceutical records including dosage and therapeutic benefits.
- Integrated image-based input for visual-to-data lookup, enriching drug records and enabling faster pharmaceutical data queries.

License Plate Detection · Python · OpenCV · Tesseract OCR *Team Project*

- Built automatic vehicle ANPR pipeline using contour detection and OCR, achieving ~90% text extraction accuracy on real-world images.
- Optimized preprocessing (thresholding, noise reduction) for robust detection across varied lighting and environmental conditions.

Aura Companion — AI Support App · Python · Web APIs · NLP · Voice *Team Project*

- Developed AI-powered web app combining voice chat and automated call-based reminders to support users experiencing social isolation.
- Implemented daily check-in scheduling workflows improving consistent user engagement through call-based notifications.

Sleep Pattern Analysis · Python · Pandas · Scikit-learn *Team Project*

- Analysed multi-stage sleep data (light/deep/REM) across 1,000+ records; built 3 classifiers (LR, RF, SVM) to predict optimal wake windows.
- Random Forest model achieved best F1-score of 0.83; visualised stage distribution with Matplotlib and Seaborn.

RESEARCH

CoT Framework for Drug Interpretation via LLMs · NLP · Biomedical AI *In Progress*

- Developing Chain-of-Thought prompting strategy to guide LLMs through multi-step drug interaction reasoning for clinical decision support.
- Combines biomedical NLP and predictive modelling to improve drug outcome interpretability; paper co-authored with faculty.

INTERNSHIP EXPERIENCE

Technology Intern (Onsite) · Evonsys, Chennai *Jun – Jul 2025*

- Trained in Pega low-code enterprise development; mapped 10+ business process requirements to application modules.
- Documented workflow specifications across 3 project modules, reducing requirement ambiguity for the development team.

INTERESTS

- Database Security
- Competitive Programming
- Classical Dance (Natya Padmam)
- Painting & Craftworks

LEADERSHIP & EXTRACURRICULARS

- Co-Secretary — ISOI Chapter · VIT Vellore · Certificate of Appreciation** 2025
- Managed chapter communications, event planning, and coordination for 50+ members over 1 year of certified service.
- VTOP Representative — ISOI Chapter · VIT Vellore · Certificate of Appreciation** 2025
- Served as official chapter representative; recognised with Certificate of Appreciation from VIT Office of Students' Welfare.
- Event Organiser — Sensor-Meha Tech Hack · VIT Vellore** 2025
- Coordinated logistics and participant onboarding for 10+ technical events with 200+ attendees.
- National Cadet Corps (NCC) ·** 2018
- Cultivated discipline, leadership, and teamwork through structured training programmes.
- Natya Padmam — Classical Dance · Formal Recognition of Excellence** Jan 2026
- Awarded prestigious title after 10+ years of dedicated training and performance in classical dance.